Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits
PEA307	ADVANCED ANALYTICAL SKILLS-I	2	1	0	3
Course Weightage	ATT: 15 CA: 30 MTT: 15 ETT: 40				

Course Focus	EMPLOYABILITY,ENTREPRENEURSHIP,SKILL DEVELOPMENT
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Course Outcomes :Through this course students should be able to

CO1 :: demonstrate procedural fluency with numerations and mathematical operations to solve the stated problems

CO2 :: use the concepts to solve various percentage and income based questions

CO3 :: analyze the data given and interpret it for solving flow chart and number series

CO4 :: employ the analytical concepts learnt to solve questions of ratio and proportion

CO5 :: use the analytical concepts learnt to solve questions of ratio and proportion

CO6 :: analyze an appropriate approach to solve logical reasoning problems related to blood relations and directions

	TextBooks (T)		
Sr No	**Title**	**Author**	**Publisher Name**
T-1	A MODERN APPROACH TO VERBAL AND NON-VERBAL REASONING	DR. R.S. AGGARWAL	S. CHAND & COMPANY
T-2	MAGICAL BOOK ON QUICKER MATHS	M.TYRA	BANKING SERVICE CHRONICLE

	Reference Books (R)		
Sr No	**Title**	**Author**	**Publisher Name**
R-1	QUANTITATIVE APTITUDE FOR COMPETITIVE EXAMINATIONS	DR.R.S. AGGARWAL	S. CHAND & COMPANY
R-2	MAGICAL BOOK SERIES ANALYTICAL REASONING	M.K. PANDEY	BANKING SERVICE CHRONICLE

Relevant Websites (RW)		
Sr No	**(Web address) (only if relevant to the course)**	**Salient Features**
RW-1	https://www.indiabix.com/	Aptitude question practice
RW-2	https://www.geeksforgeeks.org	aptitude questions and answers

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LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Speed maths(fast calculation techniques)	T-2 R-1		Zero Lecture and Speed Maths	It will help the student to understand the importance of the subject, the scoring techniques and calculation techniques for this subject.	ppt and white board.	
	Lecture 2	Advanced numeration (classification of numbers)	T-1 T-2 R-1 R-2		The students will solve the questions of numeration	Able to solve the numeration related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Advanced numeration (divisibility rules)	T-2 R-1	RW-1 RW-2	The students will solve the questions of numeration	Able to solve the numeration related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	

Week 1	Lecture 2	Advanced numeration (factors)	T-2 R-1		The students will solve the questions of numeration	Able to solve the numeration related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Advanced numeration (factorials)	T-1 T-2 R-1 R-2		The students will solve the questions of numeration	Able to solve the numeration related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
Week 2	Lecture 3	Advanced numeration(unit digit and cyclicity)	T-2 R-1		The students will solve the questions of numeration	Able to solve the numeration related problems quickly so that students can solve the Placement Test/Competitive Test paper in given the time period	ppt and white board	
		Advanced numeration (remainder theorems)	T-2 R-1		The students will solve the questions of numeration	Able to solve the numeration related problems quickly so that students can solve the Placement Test/Competitive Test paper in given the time period	ppt and white board	
	Lecture 4	Advanced numeration(HCF and LCM)	T-2 R-1		The students will solve the questions of numeration	Able to solve the numeration related problems quickly so that students can solve the Placement Test/Competitive Test paper in given the time period	ppt and white board	
Week 3	Lecture 5	Mean(advance methods of mean calculation)	T-2 R-1		The students will solve the questions of mean	Able to solve the mean based problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	

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Week 3	Lecture 5	Mean(combined mean)	T-2 R-1		The students will solve the questions of mean	Able to solve the mean based problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Mean(inclusion and exclusion related problems)	T-2 R-1		The students will solve the questions of mean	Able to solve the mean based problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
	Lecture 6	Advanced percentage (percentage to fraction conversion)	T-2 R-1		The students will solve the questions of Percentage	Able to solve the Percentage related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Advanced percentage (successive percentage change)	T-2 R-1		The students will solve the questions of Percentage	Able to solve the Percentage related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
Week 4	Lecture 7	Advanced percentage (problems related to examination, election, partial percentage and percentage error)	T-2 R-1		The students will solve the questions of Percentage	Able to solve the Percentage related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	

Week 4	Lecture 8	Income based questions (concept of cost price, selling price and marked price)	T-2 R-1		The students will solve the income based questions	Able to solve the Income based problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Income based questions (successive discount)	T-2 R-1		The students will solve the income based questions	Able to solve the Income based problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
Week 5	Lecture 9	Income based questions (problems based on number of articles and dishonest seller)	T-2 R-1		The students will solve the income based questions	Able to solve the Income based problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
	Lecture 10				Test 1			
Week 6	Lecture 11	Flow chart(problems based on data flow diagrams)	T-2 R-1		The students will solve the questions of flowchart.	Able to solve the flow chart related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
	Lecture 12	Alpha numeric coding (coding and decoding)	T-1 R-2		The students will solve logical reasoning problems based on numeric coding	Able to solve the numeric coding related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	

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Week 6	Lecture 12	Alpha numeric coding (number series)	T-1 R-2		The students will solve logical reasoning problems based on numeric coding	Able to solve the numeric coding related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
Week 7	Lecture 13	Alpha numeric coding (alphabet series)	T-1 R-2		The students will solve logical reasoning problems based on numeric coding	Able to solve the numeric coding related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Alpha numeric coding (alphanumeric series)	T-1 R-2		The students will solve logical reasoning problems based on numeric coding	Able to solve the numeric coding related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Alpha numeric coding (alphabet test)	T-1 R-2		The students will solve logical reasoning problems based on numeric coding	Able to solve the numeric coding related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		SPILL OVER						
Week 7	Lecture 14				Spill Over			
		MID-TERM						

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Week 8	Lecture 15	Advanced ratio and proportion(concept of ratio and proportion)	T-2 R-1		Students solve the problems on ratio and proportion	Able to solve the ratio and proportion related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
	Lecture 16	Advanced ratio and proportion(problems based on ages and partnership)	T-2 R-1		Students solve the problems ratio and proportion	Able to solve the ratio and proportion related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
Week 9	Lecture 17	Components and blending (concept of alligation and mixtures)	T-2 R-1		Students will solve the problems based on components and blending	Able to solve the components and blending related problems quickly so that students can solve the Placement Test/Competitive Test paper in the given time	ppt and white board	
	Lecture 18	Components and blending (problems based on removal of mixture)	T-2 R-1		Students will solve the problems based on components and blending	Able to solve the components and blending related problems quickly so that students can solve the Placement Test/Competitive Test paper in the given time	ppt and white board	
Week 10	Lecture 19				Test 2			
	Lecture 20	Counting methods(principles of counting)	T-2 R-1		Students will solve the problems on counting methods	Able to solve the problems related to counting so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	

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Week 10	Lecture 20	Counting methods (numerical permutation (formation of numbers and sum of numbers))	T-2 R-1		Students will solve the problems on counting methods	Able to solve the problems related to counting so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Counting methods(alpha permutation(rearrangement of words and rank of a word))	T-2 R-1		Students will solve the problems on counting methods	Able to solve the problems related to counting so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
Week 11	Lecture 21	Counting methods(circular arrangement)	T-2 R-1		Students will solve the problems on counting methods	Able to solve the problems related to counting so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Counting methods (distribution based questions)	T-2 R-1		Students will solve the problems on counting methods	Able to solve the problems related to counting so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Counting methods(formation of committee)	T-2 R-1		Students will solve the problems on counting methods	Able to solve the problems related to counting so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Counting methods(geometry based problems)	T-2 R-1		Students will solve the problems on counting methods	Able to solve the problems related to counting so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	

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Week 11	Lecture 22	Probability(concept of probability)	T-2 R-1		Students solve the problems based on probability	Able to solve the Probability related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Probability(classification of events)	T-2 R-1		Students solve the problems based on probability	Able to solve the Probability related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Probability(problems based on coins dices and cards)	T-2 R-1		Students solve the problems based on probability	Able to solve the Probability related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
Week 12	Lecture 23	Probability(conditional probability)	T-2 R-1		Students solve the problems based on probability	Able to solve the Probability related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
	Lecture 24	Interest based problems (concepts of interest calculations)	T-1 T-2 R-1 R-2		The students will solve the Interest based questions.	Able to solve the Interest based problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	

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Week 12	Lecture 24	Interest based problems (difference between CI and SI)	T-1 T-2 R-1 R-2		The students will solve the Interest based questions.	Able to solve the Interest based problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Interest based problems(EMI calculations)	T-1 T-2 R-1 R-2		The students will solve the Interest based questions.	Able to solve the Interest based problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
Week 13	Lecture 25	Analytical reasoning (Direction Sense)	T-1 R-2		Students solve the problems based Analytical reasoning	Able to solve the analytical reasoning related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Analytical reasoning(Blood Relation)	T-1 R-2		Students solve the problems based Analytical reasoning	Able to solve the analytical reasoning related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
	Lecture 26	Non verbal reasoning(visual reasoning: mirror/water image)	T-1 R-2		Students will solve problems based on non verbal reasoning	Able to solve the non verbal reasoning related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	

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Week 13	Lecture 26	Non verbal reasoning(paper cutting and folding)	T-1 R-2		Students will solve problems based on non verbal reasoning	Able to solve the non verbal reasoning related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Non verbal reasoning (completion of the figure)	T-1 R-2		Students will solve problems based on non verbal reasoning	Able to solve the non verbal reasoning related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Non verbal reasoning (embedded figure)	T-1 R-2		Students will solve problems based on non verbal reasoning	Able to solve the non verbal reasoning related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
		Non verbal reasoning (deviation of figure)	T-1 R-2		Students will solve problems based on non verbal reasoning	Able to solve the non verbal reasoning related problems quickly so that students can solve the Placement Test/Competitive Test paper in given time period	ppt and white board	
Week 14	Lecture 27				Test 3			
		SPILL OVER						
Week 14	Lecture 28				Spill Over			
Week 15	Lecture 29				Spill Over			
	Lecture 30				Spill Over			

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Scheme for CA:

CA Category of this Course Code is:C010203 (Total 4 tasks, 1 compulsory and out of remaining 2 best out of 3 to be considered)

Component	Weightage (%)	Mapped CO(s)
Test 1	50	CO1, CO2
Test 2	50	CO3, CO4
Test 3	50	CO5, CO6

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Test 1	To test the ability of the students to clear the cut off of the industry	30 MCQ type questions would be asked covering topics from Lecture No. 1 to 9, Duration of the first CA is 40 minutes and 25 negative marking will be imposed for every incorrect answer	Individual	Online	30	4 / 5
Test 2	To test the ability of the students to clear the cut off of the industry	30 MCQ type questions would be asked covering topics from Lecture No. 11 to 18, Duration of the second CA is 40 minutes and 25 % negative marking will be imposed for every incorrect answer	Individual	Online	30	9 / 10
Test 3	To test the ability of the students to clear the cut off of the industry	30 MCQ type questions would be asked covering topics from Lecture No. 20 to 26, Duration of the third CA is 40 minutes and 25% negative marking will be imposed for every incorrect answer	Individual	Online	30	13 / 14

Plan for Tutorial: (Please do not use these time slots for syllabus coverage)

Tutorial No.	Lecture Topic	Type of pedagogical tool(s) planned (case analysis,problem solving test,role play,business game etc)
Tutorial1	Practice questions on advance numeration	Problem Solving
Tutorial2	Practice questions on mean	Problem Solving
Tutorial3	Practice questions on advance percentage	Problem Solving

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Tutorial4	Practice questions based on income	Problem Solving
Tutorial5	Practice questions based on income	Problem Solving
Tutorial6	Practice questions based on flowchart	Problem Solving
Tutorial7	Practice questions on alpha numeric coding	Problem Solving
After Mid-Term		
Tutorial8	Practice questions on advanced ratio and proportion	Problem Solving
Tutorial9	Practice questions on components and blending	Problem Solving
Tutorial10	Practice questions on counting methods	Problem Solving
Tutorial11	Practice questions on probability	Problem Solving
Tutorial12	Practice questions based on interest	Problem Solving
Tutorial13	Practice questions on direction sense and blood relations	Problem Solving
Tutorial14	Practice questions on non verbal reasoning	Problem Solving

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